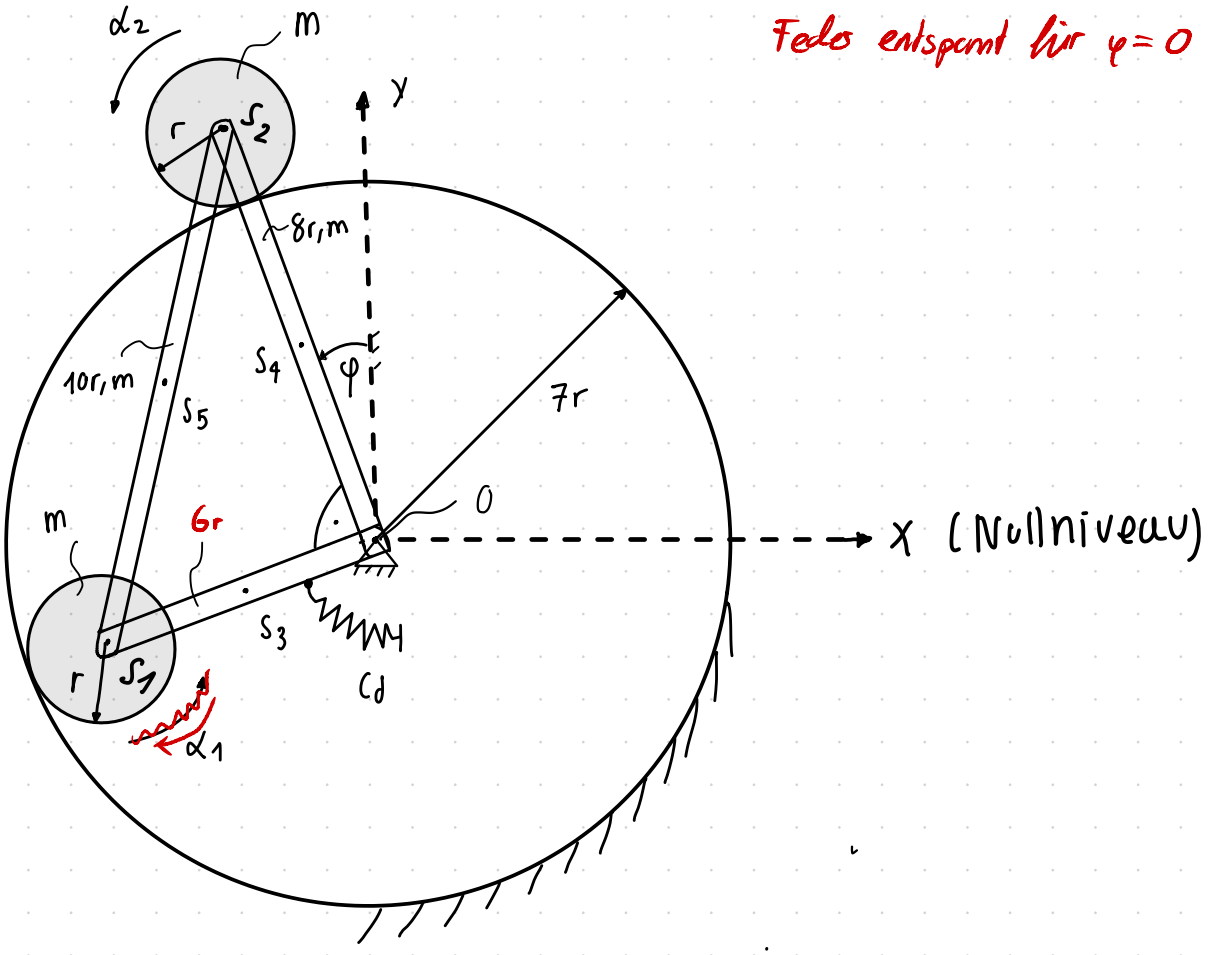
Aufg. 1 Energieerhaltungssatz



Feder entspannt für $\varphi = 0$

a) $\underline{\Gamma}_{S1}, \underline{\Gamma}_{S2}, \underline{\Gamma}_{S3}, \underline{\Gamma}_{S4}, \underline{\Gamma}_{S5}, \underline{\dot{\Gamma}}_{S1}, \underline{\dot{\Gamma}}_{S2}, \underline{\dot{\Gamma}}_{S3}, \underline{\dot{\Gamma}}_{S4}, \underline{\dot{\Gamma}}_{S5}$

b) $\dot{\alpha}_1(\varphi), \dot{\alpha}_2(\varphi)$

c) θ_0^{Δ}

d) T und U

e) $\dot{\varphi}^2(\varphi)$ mit Energieerhaltungssatz

4)

a) $\underline{r}_{S1} = \begin{pmatrix} -6r\cos\varphi \\ -6r\sin\varphi \end{pmatrix}$, $\underline{\dot{r}}_{S1} = \begin{pmatrix} 6r\dot\varphi\sin\varphi \\ -6r\dot\varphi\cos\varphi \end{pmatrix}$

$\underline{r}_{S2} = \begin{pmatrix} -8r\sin\varphi \\ 8r\cos\varphi \end{pmatrix}$, $\underline{\dot{r}}_{S2} = \begin{pmatrix} -8r\dot\varphi\cos\varphi \\ -8r\dot\varphi\sin\varphi \end{pmatrix}$

$\underline{r}_{S3} = \begin{pmatrix} -3r\cos\varphi \\ -3r\sin\varphi \end{pmatrix}$, $\underline{\dot{r}}_{S3} = \begin{pmatrix} 3r\dot\varphi\sin\varphi \\ -3r\dot\varphi\cos\varphi \end{pmatrix}$

$\underline{r}_{S4} = \begin{pmatrix} -4r\sin\varphi \\ 4r\cos\varphi \end{pmatrix}$, $\underline{\dot{r}}_{S4} = \begin{pmatrix} -4r\dot\varphi\cos\varphi \\ -4r\dot\varphi\sin\varphi \end{pmatrix}$

$\underline{r}_{S5} = \underline{r}_{S4} + \underline{r}_{S3} = \begin{pmatrix} -4r\sin\varphi - 3r\cos\varphi \\ 4r\cos\varphi - 3r\sin\varphi \end{pmatrix}$, $\underline{\dot{r}}_{S5} = \begin{pmatrix} -4r\dot\varphi\cos\varphi + 3r\dot\varphi\sin\varphi \\ -4r\dot\varphi\sin\varphi - 3r\dot\varphi\cos\varphi \end{pmatrix}$

b)

$\underline{v}_{S1} = \underline{0} + \begin{pmatrix} 0 \\ 0 \\ \dot\alpha_1 \end{pmatrix} \times \begin{pmatrix} r\cos\varphi \\ r\sin\varphi \\ 0 \end{pmatrix} = \begin{pmatrix} -\dot\alpha_1 r\sin\varphi \\ \dot\alpha_1 r\cos\varphi \\ 0 \end{pmatrix}$

$\underline{v}_{S1} = \underline{0} + \begin{pmatrix} 0 \\ 0 \\ \dot\varphi \end{pmatrix} \times \begin{pmatrix} -6r\cos\varphi \\ -6r\sin\varphi \\ 0 \end{pmatrix} = \begin{pmatrix} 6r\dot\varphi\sin\varphi \\ -6r\dot\varphi\cos\varphi \\ 0 \end{pmatrix}$

$\Rightarrow \dot\alpha_1 r = 6r\dot\varphi \Rightarrow \dot\alpha_1 = 6\dot\varphi$

$\underline{v}_{S2} = \underline{0} + \begin{pmatrix} 0 \\ 0 \\ \dot\alpha_2 \end{pmatrix} \times \begin{pmatrix} -r\sin\varphi \\ r\cos\varphi \\ 0 \end{pmatrix} = \underline{\dot r}_{S2}$

$\Rightarrow -\dot\alpha_2 r = -8r\dot\varphi \Rightarrow \dot\alpha_2 = 8\dot\varphi$

c)

$\Theta_{S5} = \frac{1}{12} m \cdot 100r^2 \Rightarrow \Theta_{S5}^0 = \frac{100}{12} mr^2 + m(16r^2 + 9r^2) = \left(\frac{25}{3} + 25\right) mr^2$

$\Theta_{S3} = \frac{1}{12} m \cdot 36r^2 \Rightarrow \Theta_{S3}^0 = 3mr^2 + 9mr^2 = 12mr^2$

$\Theta_{S4} = \frac{1}{12} m \cdot 64r^2 \Rightarrow \Theta_{S4}^0 = \frac{16}{3} mr^2 + 16mr^2 = \frac{64}{3} mr^2$

$\Rightarrow \Theta_0^A = \frac{100}{3} mr^2 + 12mr^2 + \frac{64}{3} mr^2 = \frac{200}{3} mr^2$

d)

$T = \frac{1}{2}\Theta_0^A \dot\varphi^2 + \frac{1}{2} m \dot r_{S1}^2 + \frac{1}{2}\left(\frac{1}{2} mr^2\right)\dot\alpha_1^2 + \frac{1}{2} m \dot r_{S2}^2 + \frac{1}{2}\left(\frac{1}{2} mr^2\right)\dot\alpha_2^2$

$= \frac{200}{3} mr^2\dot\varphi^2 + 18mr^2\dot\varphi^2 + \frac{1}{4}\cdot mr^2 \cdot 36\dot\varphi^2 + 32mr^2\dot\varphi^2 + \frac{1}{4}\cdot mr^2 \cdot 64\dot\varphi^2$

$= \left(\frac{200}{6} + 18 + 9 + 32 + 16\right) mr^2\dot\varphi^2 = \left(\frac{100}{3} + \frac{225}{3}\right) mr^2\dot\varphi^2 = \frac{325}{3} mr^2\dot\varphi^2$

$U = -mgr3\sin\varphi - 6mgr\sin\varphi + mgr\cdot 4\cos\varphi + 8mgr\cos\varphi + mgr(4r\cos\varphi - 3r\sin\varphi) + \frac{1}{2} c_d \varphi^2$

$= -12mgr\sin\varphi + 16mgr\cos\varphi + \frac{1}{2} c_d \varphi^2$

e) $T_0 = 0$, $T_1 = \frac{325}{3} mr^2\dot\varphi^2$, $U_0 = 16mgr$

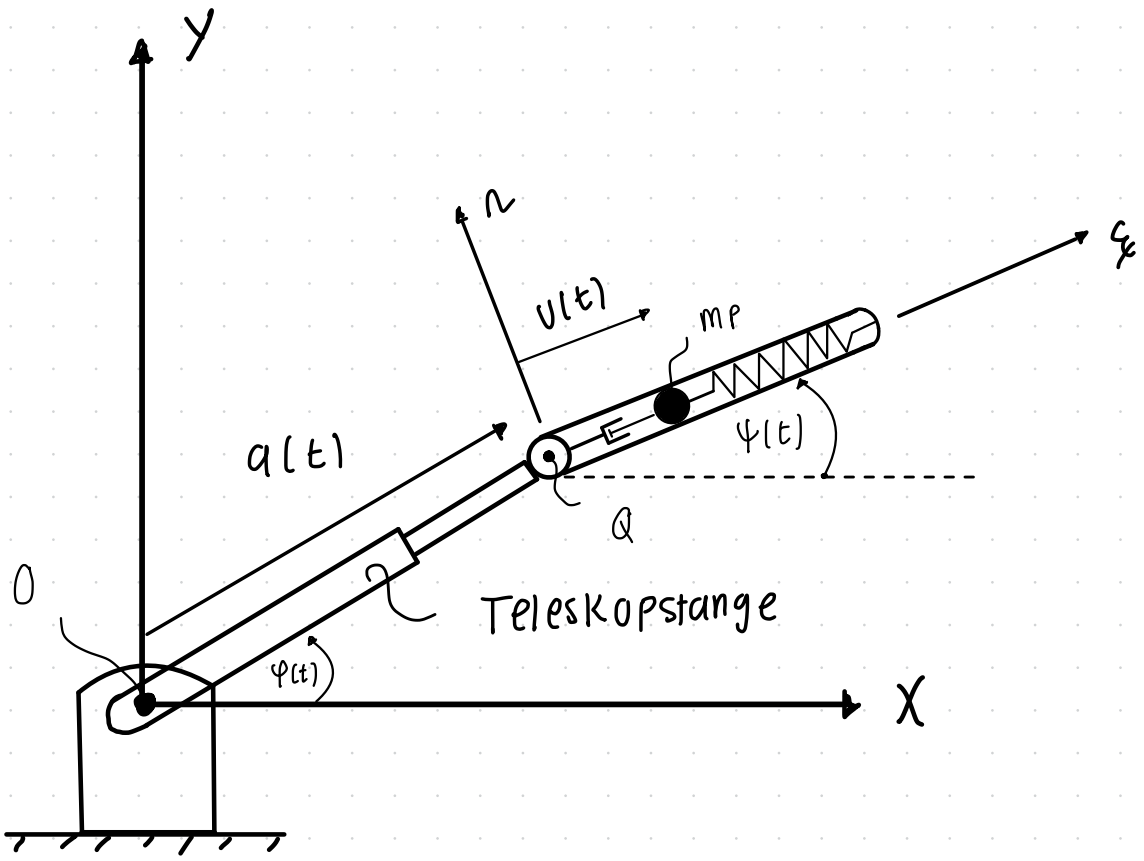
$\Rightarrow \frac{325}{3} mr^2\dot\varphi^2 - 12mgr\sin\varphi + 16mgr\cos\varphi + \frac{1}{2} c_d \varphi^2 = 16mgr$

$\Rightarrow \dot\varphi^2 = \frac{3}{325}\left[\frac{16mgr(1-\cos\varphi) + 12mgr\sin\varphi - \frac{1}{2} c_d \varphi^2}{mr^2}\right]$

$\Rightarrow \dot\varphi = \frac{1}{r}\left[\frac{3}{325}\left(16gr(1-\cos\varphi) + 12gr\sin\varphi - \frac{c_d}{m}\varphi^2\right)\right]^{1/2}$

Aufg. 2 Relativmechanik

Feder entspannt für $u = u_0$



a) $\underline{r}_Q$, $\underline{v}_Q$, $\underline{a}_Q$ im ξ, η, ζ-System

b) $\underline{P}_{QP}$, $\underline{v}_{führ}$, $\underline{v}_{rel}$, $\underline{v}$ im ξ, η, ζ-System

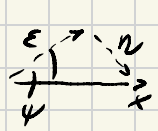
c) $\underline{a}_{führ}$, $\underline{a}_{rel}$, $\underline{a}_{cor}$, $\underline{a}$ im ξ, η, ζ-System

d) mp freischneiden, Schwerpunktsatz in ξ, η Richtung

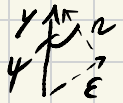
2)

a) $\underline{r}_Q = a(t)\cos\varphi\,\underline{e}_x + a(t)\sin\varphi\,\underline{e}_y$, $\underline{e}_x = \cos\psi\,\underline{e}_\xi - \sin\psi\,\underline{e}_\eta$

$\underline{e}_y = \sin\psi\,\underline{e}_\xi + \cos\psi\,\underline{e}_\eta$



$\Rightarrow \underline{r}_Q = (a(t)\cos\varphi\cos\psi + a(t)\sin\varphi\sin\psi)\,\underline{e}_\xi + (-a(t)\cos\varphi\sin\psi + a(t)\sin\varphi\cos\psi)\,\underline{e}_\eta$



$\underline{v}_Q = (\dot a(t)\cos\varphi - a(t)\dot\varphi\sin\varphi)\,\underline{e}_x + (\dot a(t)\sin\varphi + a(t)\dot\varphi\cos\varphi)\,\underline{e}_y$

$= (\dot a(t)\cos\varphi\cos\psi - a(t)\dot\varphi\sin\varphi\cos\psi + \dot a(t)\sin\varphi\sin\psi + a(t)\dot\varphi\cos\varphi\sin\psi)\,\underline{e}_\xi$
$+ (-\dot a(t)\cos\varphi\sin\psi + a(t)\dot\varphi\sin\varphi\sin\psi + \dot a(t)\sin\varphi\cos\psi + a(t)\dot\varphi\cos\varphi\cos\psi)\,\underline{e}_\eta$

ξ: $\dot a(t)\,(\cos(\varphi-\psi)) + a(t)\dot\varphi\,(-\sin(\varphi-\psi) - \sin(\varphi+\psi) + \sin(\psi-\varphi) + \sin(\varphi+\psi))$

$= \dot a(t)\cos(\varphi-\psi) - a(t)\dot\varphi\sin(\varphi-\psi)$

η: $\dot a(t)\sin(\varphi-\psi) + a(t)\dot\varphi\cos(\varphi-\psi)$

$\Rightarrow \underline{v}_Q = \begin{pmatrix} \dot a\cos(\varphi-\psi) - a\dot\varphi\sin(\varphi-\psi) \\ \dot a\sin(\varphi-\psi) + a\dot\varphi\cos(\varphi-\psi) \end{pmatrix}$

Additionstheoreme

$$\begin{aligned} \sin(a)\cos(b) &= \tfrac{1}{2}[\sin(a-b)+\sin(a+b)] \\ \sin(a)\sin(b) &= \tfrac{1}{2}[\cos(a-b)-\cos(a+b)] \\ \cos(a)\cos(b) &= \tfrac{1}{2}[\cos(a-b)+\cos(a+b)] \end{aligned}$$

$\underline{a}_Q = (\ddot a\cos\varphi - 2\dot a\dot\varphi\sin\varphi - a\ddot\varphi\sin\varphi - a\dot\varphi^2\cos\varphi)\,\underline{e}_x$
$+ (\ddot a\sin\varphi + 2\dot a\dot\varphi\cos\varphi + a\ddot\varphi\cos\varphi - a\dot\varphi^2\sin\varphi)\,\underline{e}_y$

$= (\ddot a\cos\varphi\cos\psi - 2\dot a\dot\varphi\sin\varphi\cos\psi - a\ddot\varphi\sin\varphi\cos\psi - a\dot\varphi^2\cos\varphi\cos\psi$
$+ \ddot a\sin\varphi\sin\psi + 2\dot a\dot\varphi\cos\varphi\sin\psi + a\ddot\varphi\cos\varphi\sin\psi - a\dot\varphi^2\sin\varphi\sin\psi)\,\underline{e}_\xi$
$+ (-\ddot a\cos\varphi\sin\psi + 2\dot a\dot\varphi\sin\varphi\sin\psi + a\ddot\varphi\sin\varphi\sin\psi + a\dot\varphi^2\cos\varphi\sin\psi$
$+ \ddot a\sin\varphi\cos\psi + 2\dot a\dot\varphi\cos\varphi\cos\psi + a\ddot\varphi\cos\varphi\cos\psi - a\dot\varphi^2\sin\varphi\cos\psi)\,\underline{e}_\eta$

ξ: $\ddot a\cos(\varphi-\psi) - 2\dot a\dot\varphi\sin(\varphi-\psi) - a\ddot\varphi\sin(\varphi-\psi) - a\dot\varphi^2\cos(\varphi-\psi)$

η: $\ddot a\sin(\varphi-\psi) + 2\dot a\dot\varphi\cos(\varphi-\psi) + a\ddot\varphi\cos(\varphi-\psi) - a\dot\varphi^2\sin(\varphi-\psi)$

$\Rightarrow \underline{a}_Q = \begin{bmatrix} \ddot a\cos(\varphi-\psi) - 2\dot a\dot\varphi\sin(\varphi-\psi) - a\ddot\varphi\sin(\varphi-\psi) - a\dot\varphi^2\cos(\varphi-\psi) \\ \ddot a\sin(\varphi-\psi) + 2\dot a\dot\varphi\cos(\varphi-\psi) + a\ddot\varphi\cos(\varphi-\psi) - a\dot\varphi^2\sin(\varphi-\psi) \end{bmatrix}$

b)

$\underline{s}_{QP} = \begin{pmatrix} u \\ 0 \\ 0 \end{pmatrix} = u\,\underline{e}_\xi$

$\underline{v}_{für} = \underline{v}_Q + \begin{pmatrix} 0 \\ 0 \\ \dot\psi \end{pmatrix} \times \begin{pmatrix} u \\ 0 \\ 0 \end{pmatrix} = \underline{v}_Q + \begin{pmatrix} 0 \\ \dot\psi u \\ 0 \end{pmatrix} = \begin{bmatrix} \dot a\cos(\varphi-\psi) - a\dot\varphi\sin(\varphi-\psi) \\ \dot a\sin(\varphi-\psi) + a\dot\varphi\cos(\varphi-\psi) + \dot\psi u \\ 0 \end{bmatrix}$

$\underline{v}_{rel} = \frac{d'}{dt}\underline{s}_{QP} = \dot u\,\underline{e}_\xi$

$\Rightarrow \underline{v} = \begin{bmatrix} \dot a\cos(\varphi-\psi) - a\dot\varphi\sin(\varphi-\psi) + \dot u \\ \dot a\sin(\varphi-\psi) + a\dot\varphi\cos(\varphi-\psi) + \dot\psi u \\ 0 \end{bmatrix}$

c)

$\underline{a}_{für} = \underline{a}_Q + \begin{pmatrix} 0 \\ 0 \\ \ddot\psi \end{pmatrix} \times \begin{pmatrix} u \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \dot\psi \end{pmatrix} \times \begin{pmatrix} 0 \\ \dot\psi u \\ 0 \end{pmatrix} = \underline{a}_Q + \begin{pmatrix} 0 \\ \ddot\psi u \\ 0 \end{pmatrix} + \begin{pmatrix} -\dot\psi^2 u \\ 0 \\ 0 \end{pmatrix} = \underline{a}_Q + \begin{pmatrix} -\dot\psi^2 u \\ \ddot\psi u \\ 0 \end{pmatrix}$

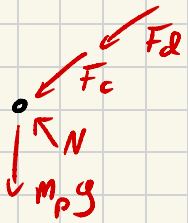
$\Rightarrow \underline{a}_{für} = \begin{bmatrix} \ddot a\cos(\varphi-\psi) - 2\dot a\dot\varphi\sin(\varphi-\psi) - a\ddot\varphi\sin(\varphi-\psi) - a\dot\varphi^2\cos(\varphi-\psi) - \dot\psi^2 u \\ \ddot a\sin(\varphi-\psi) + 2\dot a\dot\varphi\cos(\varphi-\psi) + a\ddot\varphi\cos(\varphi-\psi) - a\dot\varphi^2\sin(\varphi-\psi) + \ddot\psi u \\ 0 \end{bmatrix}$

$$\underline{a}_{rel} = \frac{d}{dt}\underline{v}_{rel} = \ddot{u}\,\underline{e}_\varepsilon$$

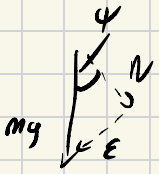
$$\underline{a}_{cor} = 2\cdot\begin{pmatrix}0\\0\\\dot\psi\end{pmatrix}\times\begin{pmatrix}\dot u\\0\\0\end{pmatrix} = 2\cdot\begin{pmatrix}0\\\dot\psi\dot u\\0\end{pmatrix} = 2\dot\psi\dot u\,\underline{e}_\eta$$

$$\Rightarrow \underline{a} = \begin{bmatrix} \ddot a\cos(\varphi-\psi) - 2\dot a\dot\varphi\sin(\varphi-\psi) - a\ddot\varphi\sin(\varphi-\psi) - a\dot\varphi^2\cos(\varphi-\psi) - \dot\psi^2 u + \ddot u \\ \ddot a\sin(\varphi-\psi) + 2\dot a\dot\varphi\cos(\varphi-\psi) + a\ddot\varphi\cos(\varphi-\psi) - a\dot\varphi^2\sin(\varphi-\psi) + \ddot\psi u + 2\dot\psi\dot u \\ 0 \end{bmatrix}$$

d)



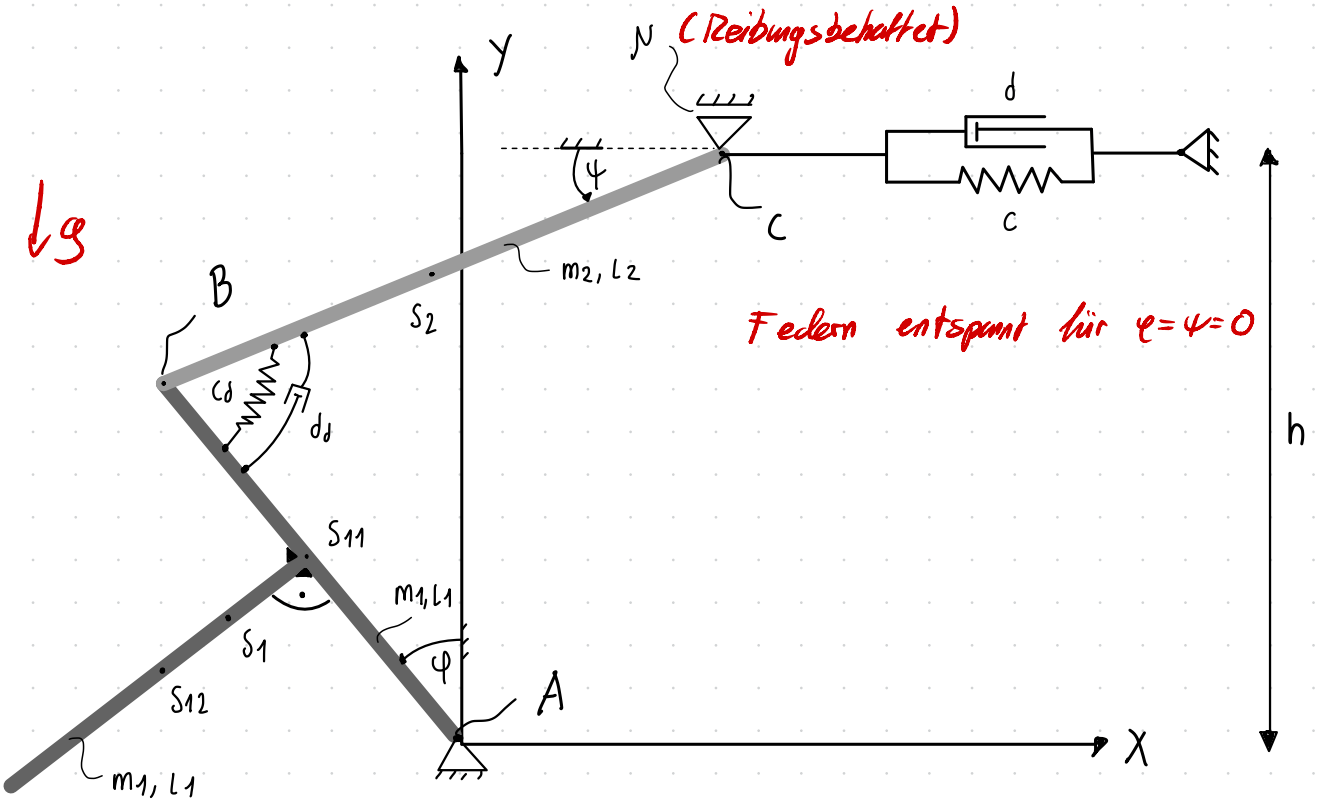
$$F_c = c\cdot(u-u_0)$$
$$F_d = d\cdot\dot u$$



ε: $m\ddot u_\varepsilon = -c(u-u_0) - d\dot u - m_p g\sin\psi$

N: $m\ddot u_\eta = N - m_p g\cos\psi$

Aufg. 3 Schwerpunktsatz / Drallsatz



lg

a) $\underline{r}_{S1}, \underline{r}_{S2}, \dot{\underline{r}}_{S1}, \dot{\underline{r}}_{S2}, \ddot{\underline{r}}_{S1}, \ddot{\underline{r}}_{S2}$

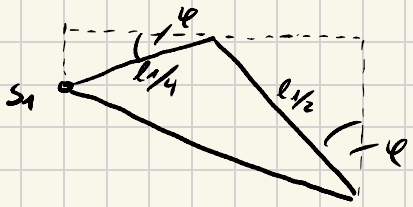
b) $\varphi(\psi)$

c) T-förmigen Körper und Stange freischneiden

d) Schwerpunktsatz in x und y Richtung sowie Drallsatz bzgl S_1 und S_2 Für beide Körper

③

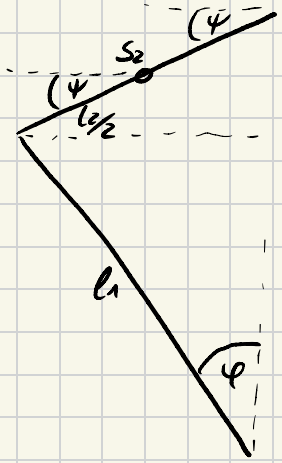
a) $\underline{r}_{S1} = \begin{pmatrix} -\frac{l_1}{2}\sin\varphi - \frac{l_1}{4}\cos\varphi \\ \frac{l_1}{2}\cos\varphi - \frac{l_1}{4}\sin\varphi \end{pmatrix}$



$\underline{r}_{S2} = \begin{pmatrix} -l_1\sin\varphi + \frac{l_2}{2}\cos\psi \\ l_1\cos\varphi + \frac{l_2}{2}\sin\psi \end{pmatrix}$

$\underline{\dot{r}}_{S1} = \begin{pmatrix} -\frac{l_1}{2}\dot\varphi\cos\varphi + \frac{l_1}{4}\dot\varphi\sin\varphi \\ -\frac{l_1}{2}\dot\varphi\sin\varphi - \frac{l_1}{4}\dot\varphi\cos\varphi \end{pmatrix}$

$\underline{\dot{r}}_{S2} = \begin{pmatrix} -l_1\dot\varphi\cos\varphi - \frac{l_2}{2}\dot\psi\sin\psi \\ -l_1\dot\varphi\sin\varphi + \frac{l_2}{2}\dot\psi\cos\psi \end{pmatrix}$



$\underline{\ddot{r}}_{S1} = \begin{pmatrix} -\frac{l_1}{2}\ddot\varphi\cos\varphi + \frac{l_1}{2}\dot\varphi^2\sin\varphi + \frac{l_1}{4}\ddot\varphi\sin\varphi + \frac{l_1}{4}\dot\varphi^2\cos\varphi \\ -\frac{l_1}{2}\ddot\varphi\sin\varphi - \frac{l_1}{2}\dot\varphi^2\cos\varphi - \frac{l_1}{4}\ddot\varphi\cos\varphi + \frac{l_1}{4}\dot\varphi^2\sin\varphi \end{pmatrix}$

$= \begin{pmatrix} -\frac{l_1}{2}(\ddot\varphi\cos\varphi - \dot\varphi^2\sin\varphi) + \frac{l_1}{4}(\ddot\varphi\sin\varphi + \dot\varphi^2\cos\varphi) \\ -\frac{l_1}{2}(\ddot\varphi\sin\varphi + \dot\varphi^2\cos\varphi) - \frac{l_1}{4}(\ddot\varphi\cos\varphi - \dot\varphi^2\sin\varphi) \end{pmatrix}$

$\underline{\ddot{r}}_{S2} = \begin{pmatrix} -l_1\ddot\varphi\cos\varphi + l_1\dot\varphi^2\sin\varphi - \frac{l_2}{2}\ddot\psi\sin\psi - \frac{l_2}{2}\dot\psi^2\cos\psi \\ -l_1\ddot\varphi\sin\varphi - l_1\dot\varphi^2\cos\varphi + \frac{l_2}{2}\ddot\psi\cos\psi - \frac{l_2}{2}\dot\psi^2\sin\psi \end{pmatrix}$

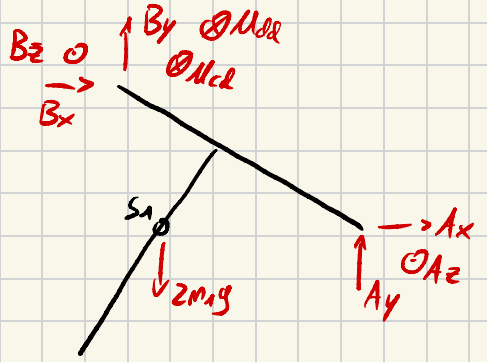
$= \begin{pmatrix} -l_1(\ddot\varphi\cos\varphi - \dot\varphi^2\sin\varphi) - \frac{l_2}{2}(\ddot\psi\sin\psi + \dot\psi^2\cos\psi) \\ -l_1(\ddot\varphi\sin\varphi + \dot\varphi^2\cos\varphi) + \frac{l_2}{2}(\ddot\psi\cos\psi - \dot\psi^2\sin\psi) \end{pmatrix}$

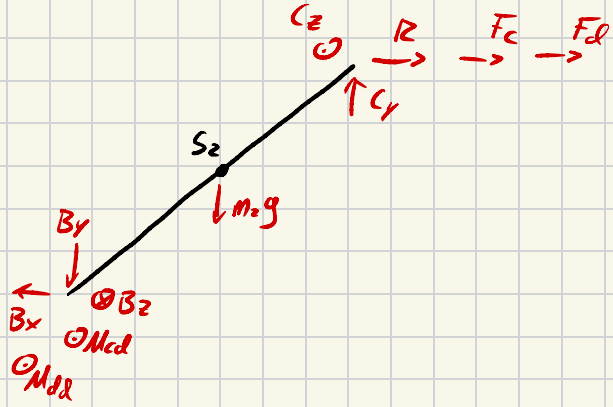
b)

$h = l_1\cos\varphi + l_2\sin\psi \quad \Rightarrow \quad \cos\varphi = \frac{h - l_2\sin\psi}{l_1}$

$\Rightarrow \varphi = \arccos\left(\frac{h - l_2\sin\psi}{l_1}\right)$

c)





d) T-Körper: x: $m\ddot r_{S1x} = A_x + B_x$, y: $m\ddot r_{S1y} = A_y - 2m_1g + B_y$, $M_d = d(\dot\psi - \dot\varphi)$, $M_c = c(\psi - \varphi)$

Θ_{S1}: $\Theta_{S1}\ddot\varphi = A_y\left(\frac{l_1}{2}\sin\varphi + \frac{l_1}{4}\cos\varphi\right) + A_x\left(\frac{l_1}{2}\cos\varphi - \frac{l_1}{4}\sin\varphi\right) - B_x\left(l_1\cos\varphi - \frac{l_2}{2}\cos\varphi + \frac{l_1}{4}\sin\varphi\right)$
$- B_y\left(l_1\sin\varphi - \frac{l_2}{2}\sin\varphi - \frac{l_1}{4}\cos\varphi\right) - M_d - M_c$
$= -M_d - M_c + A_y\left(\frac{l_1}{2}\sin\varphi + \frac{l_1}{4}\cos\varphi\right) + A_x\left(\frac{l_1}{2}\cos\varphi - \frac{l_1}{4}\sin\varphi\right) - B_x\left(\frac{l_2}{2}\cos\varphi + \frac{l_1}{4}\sin\varphi\right)$
$- B_y\left(\frac{l_2}{2}\sin\varphi - \frac{l_1}{4}\cos\varphi\right)$

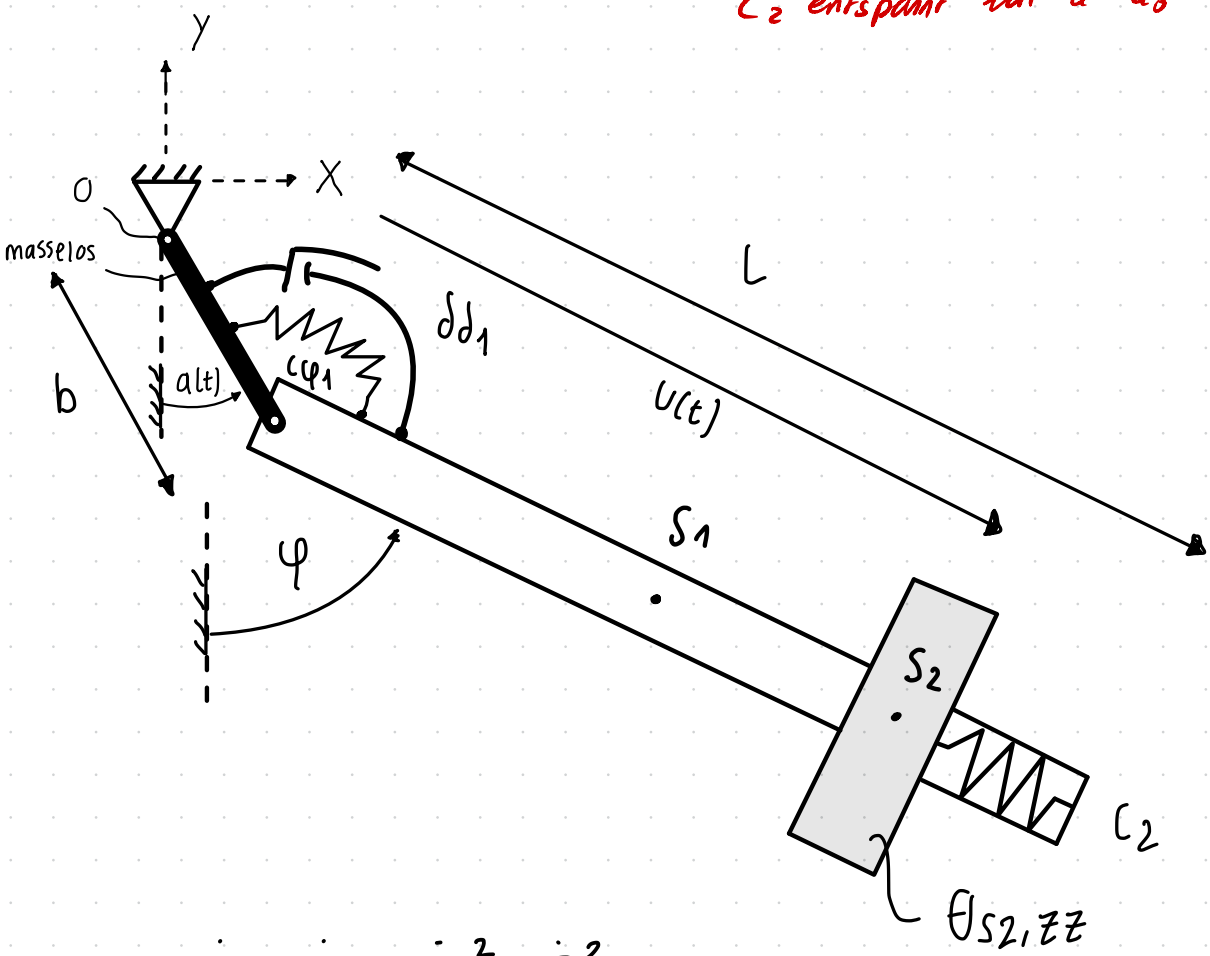
Stange: x: $m\ddot r_{S2x} = -B_x + R + c\left(l_2\cos\psi - l_2\right) + d\cdot\left(-l_2\dot\psi\sin\psi\right)$, y: $m\ddot r_{S2y} = C_y - B_y - m_2g$
$= -B_x + R + C_y + c\left(l_2\cos\psi - l_2\right) - d\left(l_2\dot\psi\sin\psi\right)$

S_2: $\Theta_{S2}\ddot\psi = -B_x\frac{l_2}{2}\sin\psi + B_y\frac{l_2}{2}\cos\psi + M_c + M_d + C_y\frac{l_2}{2}\cos\psi - (R + F_c + F_d)\frac{l_2}{2}\sin\psi$

$\Rightarrow \Theta_{S2}\ddot\psi = -B_x\frac{l_2}{2}\sin\psi + (B_y + C_y)\frac{l_2}{2}\cos\psi + c(\psi - \varphi) + d(\dot\psi - \dot\varphi) - \left(R + c\left(l_2\cos\psi - l_2\right) - d\,l_2\dot\psi\sin\psi\right)\frac{l_2}{2}\sin\psi$

Lagrange 2 Art und Schwingungen

$c_{\varphi 1}$ entspannt für $\varphi = 0$
c_2 entspannt für $u = u_0$



a) $\underline{r}_{S1}, \underline{r}_{S2}, \dot{\underline{r}}_{S1}, \dot{\underline{r}}_{S2}, \dot{\underline{r}}_{S1}^2, \dot{\underline{r}}_{S2}^2$

b) T und U allgemeine Lage

c) Q_u^{enk}, Q_φ^{enk}

d) ab hier $\alpha = 0$, $\dot{\alpha} = 0$, $\ddot{\alpha} = 0$
Lagrange Gleichungen in u und φ.

e) Gegebene Schwingungs DGL, ω_0, D bestimmen und $\hat{\varphi}$

f) Irgendwas mit η_{krit}

④

a) $\underline{r}_{S1} = \begin{pmatrix} \frac{l}{2}\sin\varphi \\ -\frac{l}{2}\cos\varphi \end{pmatrix} + \begin{pmatrix} b\sin\alpha \\ -b\cos\alpha \end{pmatrix} = \begin{pmatrix} b\sin\alpha + \frac{l}{2}\sin\varphi \\ -\frac{l}{2}\cos\varphi - b\cos\alpha \end{pmatrix}$

$\underline{\dot{r}}_{S1} = \begin{pmatrix} b\dot{\alpha}\cos\alpha + \frac{l}{2}\dot{\varphi}\cos\varphi \\ b\dot{\alpha}\sin\alpha + \frac{l}{2}\dot{\varphi}\sin\varphi \end{pmatrix}$

$\dot{r}_{S1}^2 = b^2\dot{\alpha}^2 + b\dot{\alpha}\dot{\varphi}(\cos\alpha\cos\varphi + \sin\alpha\sin\varphi) + \frac{l^2}{4}\dot{\varphi}^2$

$= b^2\dot{\alpha}^2 + \frac{l^2}{4}\dot{\varphi}^2 + b\dot{\alpha}\dot{\varphi}(\cos(\alpha-\varphi))$

$\underline{r}_{S2} = \begin{pmatrix} b\sin\alpha + u\sin\varphi \\ -b\cos\alpha - u\cos\varphi \end{pmatrix}$

$\underline{\dot{r}}_{S2} = \begin{pmatrix} b\dot{\alpha}\cos\alpha + \dot{u}\sin\varphi + u\dot{\varphi}\cos\varphi \\ b\dot{\alpha}\sin\alpha - \dot{u}\cos\varphi + u\dot{\varphi}\sin\varphi \end{pmatrix}$

$\dot{r}_{S2} = \big[\, b^2\dot{\alpha}^2\cos^2\alpha + 2ub\dot{\alpha}\sin\varphi\cos\alpha + 2ub\dot{\alpha}\dot{\varphi}\cos\varphi\cos\alpha$
$+ \dot{u}^2\sin^2\varphi + 2u\dot{u}\sin\varphi\cos\varphi + u^2\dot{\varphi}^2\cos^2\varphi + b^2\dot{\alpha}^2\sin^2\alpha - 2\dot{u}b\dot{\alpha}\sin\alpha\cos\varphi$
$+ 2ub\dot{\alpha}\dot{\varphi}\sin\alpha\sin\varphi + \dot{u}^2\cos^2\varphi - 2\dot{u}u\dot{\varphi}\sin\varphi\cos\varphi + u^2\dot{\varphi}^2\sin^2\varphi \,\big]^{1/2}$

$\Rightarrow \dot{r}_{S2}^2 = b^2\dot{\alpha}^2 + \dot{u}^2 + u^2\dot{\varphi}^2 + 2\dot{u}b\dot{\alpha}(\sin\varphi\cos\alpha - \sin\alpha\cos\varphi) + 2ub\dot{\alpha}\dot{\varphi}(\sin\alpha\sin\varphi + \cos\varphi\cos\alpha)$

$= b^2\dot{\alpha}^2 + \dot{u}^2 + u^2\dot{\varphi}^2 + 2ub\dot{\alpha}\dot{\varphi}\cos(\alpha-\varphi) - 2\dot{u}b\dot{\alpha}\sin(\alpha-\varphi)$

b)

$T = \frac{1}{2}m_1\dot{r}_{S1}^2 + \frac{1}{2}\left(\frac{1}{12}m_1l^2\right)\dot{\varphi}^2 + \frac{1}{2}m_2\dot{r}_{S2}^2 + \frac{1}{2}\Theta_{S2}\dot{\varphi}^2$

$= \frac{1}{2}m_1\left(b^2\dot{\alpha}^2 + \frac{l^2}{4}\dot{\varphi}^2 + b\dot{\alpha}\dot{\varphi}(\cos(\alpha-\varphi)\right) + \frac{1}{24}m_1l^2\dot{\varphi}^2$
$+ \frac{1}{2}m_2\left(b^2\dot{\alpha}^2 + \dot{u}^2 + u^2\dot{\varphi}^2 + 2ub\dot{\alpha}\dot{\varphi}\cos(\alpha-\varphi) - 2\dot{u}\dot{\alpha}b\sin(\alpha-\varphi)\right) + \frac{1}{2}\Theta_{S2}\dot{\varphi}^2$

$= \frac{1}{2}m_1\left(b^2\dot{\alpha}^2 + \frac{l^2}{3}\dot{\varphi}^2 + b\dot{\alpha}\dot{\varphi}(\cos(\alpha-\varphi))\right)$
$+ \frac{1}{2}m_2\left(b^2\dot{\alpha}^2 + \dot{u}^2 + u^2\dot{\varphi}^2 + 2ub\dot{\alpha}\dot{\varphi}\cos(\alpha-\varphi) - 2\dot{u}\dot{\alpha}b\sin(\alpha-\varphi)\right) + \frac{1}{2}\Theta_{S2}\dot{\varphi}^2$

$U = -m_1g\left(\frac{l}{2}\cos\varphi + b\cos\alpha\right) - m_2g(b\cos\alpha + u\cos\varphi) + \frac{1}{2}c_\varphi\cdot\varphi^2 + \frac{1}{2}c(u-u_0)^2$

$= -m_1g\left(\frac{l}{2}\cos\varphi + b\cos\alpha\right) - m_2g(b\cos\alpha + u\cos\varphi) + \frac{1}{2}c_\varphi(\varphi-\alpha)^2 + \frac{c}{2}(u-u_0)^2$

c)

$\delta W = -d_{11}(\dot{\varphi}-\dot{\alpha})(\delta\varphi - \delta\alpha)$

$\Rightarrow Q_\varphi^{einh} = -d_1(\dot{\varphi}-\dot{\alpha})$

$Q_u^{einh} = 0$

d)

$U = -m_1g\left(\frac{l}{2}\cos\varphi + b\right) - m_2g(u\cos\varphi + b) + \frac{1}{2}c_\varphi\varphi^2 + \frac{1}{2}c(u-u_0)^2$

$T = \frac{1}{2}m_1\left(\frac{l^2}{3}\dot{\varphi}^2\right) + \frac{1}{2}m_2(\dot{u}^2 + u^2\dot{\varphi}^2) + \frac{1}{2}\Theta_{S2}\dot{\varphi}^2$

$\Rightarrow L = T - U = \frac{1}{6}m_1l^2\dot{\varphi}^2 + \frac{1}{2}m_2(\dot{u}^2 + u^2\dot{\varphi}^2) + \frac{1}{2}\Theta_{S2}\dot{\varphi}^2 + m_1g\left(\frac{l}{2}\cos\varphi + b\right)$
$+ m_2g(u\cos\varphi + b) - \frac{1}{2}c_\varphi\cdot\varphi^2 - \frac{1}{2}c(u-u_0)^2$

$\Rightarrow \frac{\partial L}{\partial\dot{u}} = m_2\dot{u} \quad \Rightarrow \frac{\partial}{\partial t}\left(\frac{\partial L}{\partial\dot{u}}\right) = m_2\ddot{u}$

$\frac{\partial L}{\partial u} = m_2u\dot{\varphi}^2 + m_2g\cos\varphi - c(u-u_0)$

$\Rightarrow u: \quad 0 = m_2\ddot{u} - m_2u\dot{\varphi}^2 - m_2g\cos\varphi + c(u-u_0)$

$$\frac{\partial L}{\partial \dot\varphi} = \tfrac{1}{3} m_1 l^2 \dot\varphi + m_2 u^2 \dot\varphi + \Theta_{S2}\dot\varphi$$

$$\Rightarrow \frac{d}{dt}\left(\frac{\partial L}{\partial \dot\varphi}\right) = \tfrac{1}{3} m_1 l^2 \ddot\varphi + m_2 u^2 \ddot\varphi + 2 m_2 u \dot u \dot\varphi + \Theta_{S2}\ddot\varphi$$

$$\frac{\partial L}{\partial \varphi} = -m_1 g \tfrac{l}{2}\sin\varphi - m_2 g u \sin\varphi - c_\varphi \cdot \varphi$$

$$\Rightarrow \varphi: \quad -d_\varphi(\dot\varphi - \dot\lambda) = \tfrac{1}{3} m_1 l^2 \ddot\varphi + m_2 u^2 \ddot\varphi + 2 m_2 u \dot u \dot\varphi + \Theta_{S2}\ddot\varphi + m_1 g \tfrac{l}{2}\sin\varphi + m_2 g u \sin\varphi + c_\varphi \cdot \varphi$$

e)

Gegeben: $\tfrac{1}{3} m_1 l^2 \ddot\varphi + d\dot\varphi + \tfrac{1}{2} m_1 g l \varphi = \hat d \cdot \hat\lambda$, $\lambda(t) = \hat\lambda \sin(\Omega t)$, $\dot\lambda(t) = \hat\lambda \Omega \cos(\Omega t)$

$$\omega_0 = \sqrt{\frac{c}{m}} = \sqrt{\frac{3gl}{2l^2}} = \sqrt{\frac{3}{2}\cdot\frac{g}{l}}$$

$$D = \frac{\delta}{\omega_0} = \frac{3d}{2 m_1 l^2 \sqrt{\frac{3}{2}\frac{g}{l}}} = \frac{3d}{\sqrt{6 m_1^2 g l}} = \frac{d\sqrt{6}}{2 m_1 \cdot l \sqrt{g\cdot l}}$$

$\varphi(t) = \hat\varphi(\eta)\cdot\cos(\Omega t - \psi)$ (auf GRP ist nur Amplitude gesucht)

$$\hat\varphi(\eta) = \frac{\hat d \cdot \hat\lambda \Omega}{\tfrac{1}{2} m_1 g l} \cdot \frac{1}{\sqrt{(1-\eta^2)^2 + (2D\eta)^2}}, \quad \eta = \frac{\Omega}{\omega_0}$$

$$= \frac{2\hat d \hat\lambda \Omega}{m_1 g l} \cdot \frac{1}{\sqrt{\left(1 - \frac{2\Omega^2 l}{3g}\right)^2 + \left(\frac{d\sqrt{6}}{m_1 l \sqrt{gl}} \cdot \frac{\Omega}{\sqrt{\frac{3}{2}\frac{g}{l}}}\right)^2}}$$

$$= \frac{2\hat d \hat\lambda \Omega}{m_1 g l} \cdot \frac{1}{\sqrt{\left(1 - \frac{2\Omega^2 l}{3g}\right)^2 + \left(\frac{\Omega \cdot d \cdot 2}{m_1 l g}\right)^2}}$$

f)

$$\hat\varphi = \frac{\hat d \Omega \hat a}{\omega_0^2 \cdot m} \cdot \frac{1}{\sqrt{(1-\eta^2)^2 + (2D\eta)^2}} = \frac{\hat d \hat a}{\omega_0 m} \cdot \frac{\eta}{\sqrt{(1-\eta^2)^2 + (2D\eta)^2}}$$

$$\eta_{max}: \quad \frac{\partial \hat\varphi}{\partial \eta_{max}} \stackrel{!}{=} 0 = \frac{1}{((1-\eta^2)^2 + (2D\eta)^2)^{1/2}} - \frac{\eta}{2}\frac{(-4\eta + 4\eta^3 + 8D^2\eta^2)}{((1-\eta^2)^2 + (2D\eta)^2)^{3/2}}$$

$$\Rightarrow (1-\eta^2)^2 + (2D\eta)^2 = \eta(-2\eta + 2\eta^3 + 4D^2\eta)$$

$$\Rightarrow 1 - 2\eta^2 + \eta^4 + 4D^2\eta^2 = -2\eta^2 + 2\eta^4 + 4D^2\eta^2$$

$$\Rightarrow \eta^4 = 1 \Rightarrow \eta_{max} = 1 \quad \text{also} \quad \Omega = \omega_0$$

Allgemein (ohne Herleitung):

- Bei Kraftanregung wie $m\ddot x + d\dot x + cx = \hat F \cos(\Omega t)$ → $\eta_{max} = \sqrt{1-2D^2}$
- Bei Geschwindigkeitsanregung wie $m\ddot x + d\dot x + cx = \hat d \Omega \hat\lambda \cos(\Omega t)$ → $\eta_{max} = 1$
- Bei Beschleunigungsanregung wie $m\ddot x + d\dot x + cx = \hat m \Omega^2 d \cos(\Omega t)$ → $\eta_{max} = \frac{1}{\sqrt{1-2D^2}}$